

Reasoning-Guided Whole-Slide Report Generation via Foundation-Model Multiple-Instance Learning and Deterministic Pathology-Workflow Templating

A Submission to the REG² (REG2026) MICCAI Challenge

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Abstract

We present our submission to the **REG² (REG2026) Pathologist Reasoning-Guided Report Generation Challenge** at MICCAI 2026. The task asks a system to ingest a single haematoxylin-and-eosin (H&E) whole-slide image (WSI) and emit (i) a structured chain-of-thought (CoT) reasoning graph whose edges are *(question, next_question)* pairs, (ii) the intermediate answers along that graph, and (iii) a final free-text pathology report, while additionally serving a visual-grounding interface that must distinguish tissue from background regions of interest (ROIs). Through a systematic analysis of the 11,220-slide training corpus we establish that the reasoning graph is *highly templated*: only 93 canonical questions and 191 distinct edges occur, and conditioning on the pair *(organ, #1 diagnosis)* yields $\sim 86\%$ graph purity and $\sim 92\%$ answer purity. This reframes an apparently open-ended report-generation problem as a *constrained multi-task classification* problem followed by deterministic template lookup. Our pipeline (a) tiles each no-pyramid generic-TIFF WSI with a bounded-budget foreground sampler (~ 1.4 s/slide), (b) embeds tiles with the **CONCH** pathology vision-language foundation model, (c) aggregates them with a gated-attention multiple-instance learning (MIL) head producing organ (10-way) and diagnosis (77-way bucketed) predictions, and (d) emits the modal training CoT graph, answers, and keyword-rich report for the predicted *(organ, diagnosis)* key. A lightweight Otsu tissue-fraction module serves the visual-grounding interface. An oracle analysis bounds the achievable workflow score at **0.89** given perfect *(organ, diagnosis)* prediction; our trained system attains a held-out workflow score of **0.794** (organ accuracy 0.96, diagnosis accuracy 0.69), i.e. within 0.095 of that ceiling, with the residual gap attributable almost entirely to fine-grained diagnosis classification. We detail the data analysis, architecture, high-performance-computing infrastructure (Blackwell-class GPUs, CUDA 12.8/PyTorch 2.9), training methodology, and qualitative predictions, and release all code with reproduction instructions.

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1 Introduction

Computational pathology has been transformed by self-supervised *foundation models* trained on millions of histopathology tiles, which yield transferable tile-level representations that power weakly supervised whole-slide diagnosis [2, 8, 16, 14]. In parallel, multimodal and report-generation systems attempt to translate gigapixel slides into the narrative language pathologists use in clinical practice [5]. The **REG² (REG2026)** challenge sits precisely at this intersection: rather than predicting a single label, a system must reproduce the *reasoning process* a pathologist follows—an ordered graph of questions and answers—and the final report that the process supports.

This is attractive because it makes a model’s decision process explicit and auditable, but it is deceptively difficult to evaluate: free-text reasoning admits infinite surface forms. The challenge therefore canonicalizes the reasoning into a graph of *(question, next_question)* edges with discrete answers and scores systems against the ground-truth graph, answers, and report using a weighted composite metric (Section 2). Our central observation is that this canonicalization, combined with the clinical reality that diagnostic workups are protocolized per organ, makes the target distribution *far lower-entropy than it first appears*. We quantify this directly on the training data and exploit it.

Clinical motivation. A pathology report is the end point of a structured cognitive process: the pathologist identifies the specimen organ and procedure, screens for abnormality, enumerates and grades the diagnoses, and composes a synoptic report. Systems that emit only a final label discard this process and are correspondingly hard to trust or audit. By requiring the *reasoning graph* itself, REG2026 pushes models toward outputs that mirror clinical workflow and can be inspected step by step—a prerequisite for safe deployment in diagnostic settings. Our design preserves this audibility: every emitted graph is a verbatim, human-readable clinical workup, and the one learned decision (the organ–diagnosis key) is exactly the decision a pathologist would defend.

Contributions. This report makes the following contributions:

1. **A data-driven reduction.** We show empirically that the REG2026 reasoning graph is templated (93 questions, 191 edges) and that conditioning on *(organ, #1 diagnosis)* recovers $\sim 86\%$ of graphs and $\sim 92\%$ of answers exactly, reducing the task to multi-task WSI classification plus deterministic templating.
2. **An oracle ceiling analysis.** Using held-out templating we bound the workflow score under several oracle conditions, isolating where machine learning is actually needed (organ + diagnosis) from where it is not (graph structure, report scaffolding).
3. **A practical, offline-deployable pipeline.** A bounded-budget WSI tiler, a CONCH foundation encoder, a gated-attention multi-task MIL head, a deterministic template engine, and an Otsu visual-grounding module—engineered to run within the challenge’s offline, single-GPU, 5-minute-per-case container constraints.
4. **Reproducible engineering at scale.** We document the HPC pipeline used to embed 11,220 WSIs on Blackwell-class GPUs, including the CUDA/PyTorch compatibility work required, and release all code.

2 The REG2026 Task and Evaluation

2.1 Problem definition

The input is a single H&E WSI in tiled-TIFF format. The system must serve two interfaces:

Interface A — Workflow Reasoning.

Given the WSI, output a chain-of-thought: an ordered list of steps $\{(q_i, a_i, n_i)\}$ where q_i is a

question, a_i its answer, and n_i the next question (an empty n_i terminates the graph). The induced directed edge set $\{(q_i, n_i)\}$ is the reasoning graph.

Interface B — Visual Grounding.

Given a region-of-interest (ROI) crop and a visual-context question, decide whether the ROI contains diagnostically relevant tissue or is background, and respond accordingly.

2.2 Scoring

The overall score linearly combines the two interfaces:

$$\text{Overall} = 0.70 \cdot A + 0.30 \cdot B. \quad (1)$$

The Workflow component A is itself a weighted sum of four sub-metrics:

$$A = 0.05 \cdot \underbrace{\text{BPV}}_{\text{begin-point validity}} + 0.30 \cdot \underbrace{\text{Edge-F1}}_{\text{graph edges}} + 0.25 \cdot \underbrace{\text{MESS}}_{\text{answer match}} + 0.40 \cdot \underbrace{\text{FinalReport}}_{\text{report quality}}, \quad (2)$$

with the report sub-metric dominated by keyword recall:

$$\text{FinalReport} = 0.15 (\text{ROUGE-L} + \text{BLEU}) + 0.40 \text{ keyword-Jaccard} + 0.30 \text{ embedding-cosine}. \quad (3)$$

The grounding component $B = 0.30 \cdot B_1 + 0.30 \cdot B_2 + 0.40 \cdot B_3$ rewards background rejection (B_1), input sensitivity (B_2), and tissue/background separation (B_3). Edge-F1 and MESS require the system to emit the *exact canonical* question strings: paraphrases score zero. ROUGE-L and BLEU follow the standard definitions [7, 10].

2.3 Operational constraints

Submissions are offline Docker containers evaluated on a single NVIDIA A10G GPU with ≤ 32 GB RAM and a wall-clock budget of **<5 minutes per case**. Model weights are delivered separately from the (code-only) image. These constraints strongly shape the design: the encoder and aggregator must be light enough to embed a slide and predict within minutes, and report generation cannot rely on a large external language model.

3 Related Work

3.1 Pathology foundation encoders

Self-supervised vision transformers [3] trained on large histology corpora now provide the de-facto tile representations in computational pathology, displacing ImageNet-pretrained CNNs. UNI [2] and Virchow [14] are pure-vision encoders trained with DINOv2-style self-distillation on 10^5 – 10^6 slides; Prov-GigaPath [16] adds a slide-level aggregator pretrained on real-world data; CTransPath [15] (a hybrid CNN–transformer with semantically-relevant contrastive learning) and Phikon [4] (iBOT masked-image modeling) are widely used open encoders. **CONCH** [8] differs in being a *vision–language* model: its image tower is aligned to pathology captions through CLIP-style contrastive learning [11], producing 512-dimensional embeddings that are strong both zero-shot and as frozen features for supervised heads. Table 1 compares the main options. We adopt CONCH for its compact, semantically-aligned embeddings, and validated UNI2-h as a drop-in 1536-d alternative (Section 8); both share our `embed_tiles` interface with encoder-specific normalization.

Table 1: Representative pathology tile-encoder foundation models. “Dim” is the output embedding dimension; “VL” marks vision–language (caption-aligned) pretraining.

Encoder	Backbone	Pretraining	Dim	VL
CTransPath [15]	CNN+Swin	SRCL contrastive	768	no
Phikon [4]	ViT-B	iBOT MIM	768	no
UNI [2]	ViT-L/H	DINOv2	1024/1536	no
Virchow [14]	ViT-H	DINOv2	1280	no
Prov-GigaPath [16]	ViT-G + slide	DINOv2 + MAE	1536	no
CONCH [8]	ViT-B/16	image–text contrastive	512	yes

3.2 Weakly supervised slide classification (MIL)

Because slide-level labels rarely come with pixel annotations, whole-slide prediction is typically cast as multiple-instance learning (MIL) over tiles: a slide is a *bag* of tile instances, and only the bag carries a label. Early approaches used max/mean pooling or instance-level top- k selection [1], which first demonstrated clinical-grade performance at scale. Attention-based MIL (ABMIL) [6] replaced fixed pooling with a learned, permutation-invariant attention over instances, and its *gated* variant—which we adopt—adds a sigmoid gating branch to modulate the tanh attention, improving expressivity. CLAM [9] augments ABMIL with instance-level clustering constraints and strong data efficiency, and TransMIL [12] models inter-tile correlations with self-attention rather than treating tiles as independent. We use a gated-attention head for its simplicity, speed, and the interpretable attention map it yields (useful for the qualitative analysis in Section 10); the multi-task extension to two heads is, to our knowledge, the natural fit for the (organ, diagnosis) factorization the REG2026 metric rewards.

3.3 Histopathology report generation

Report-generation systems such as HistGen [5] encode local–global slide features and decode narrative text with cross-modal context interaction; broader surveys [13] catalogue the rapid growth of deep learning in histopathology. Such free-form decoders are expressive but computationally heavy and difficult to evaluate, and they are poorly matched to an offline, five-minute, single-GPU container. Crucially, the REG2026 report sub-metric is *keyword-Jaccard-dominated* and the workflow component is *graph-structured with exact-string matching*, both of which reward *content coverage* over fluent prose. This, plus the compute budget, motivates our template-driven assembly over a learned decoder: a correctly-keyed template reproduces the reasoning graph exactly and supplies the diagnostic keywords the metric scores, at negligible cost.

3.4 Reasoning structure and evaluation of generated text

Framing model outputs as an explicit graph of questions and answers connects REG2026 to chain-of-thought and structured-reasoning literature, but with a decisive twist: the reasoning space is *closed and protocolized* (93 questions, 191 edges; Section 4) rather than open-domain, so exact graph reconstruction is both well-defined and attainable by lookup. Text-overlap metrics ROUGE-L [7] and BLEU [10] contribute only 0.15 of the report sub-score, while keyword recall and embedding similarity dominate—a weighting that, as our oracle analysis shows (Section 9), leaves near-zero headroom for learned generation beyond correct templating.

4 Dataset and Empirical Analysis

4.1 Composition

The training set comprises **11,220** H&E WSIs, each paired with a ground-truth chain-of-thought in `train_CoT.json`. A separate `test_phase1` split provides 350 slides for leaderboard evaluation, plus a small `debug` split. The slides are single-level, tiled (256 px) JPEG-compressed *generic* TIFFs *without* an image pyramid—a property with major engineering consequences (Section 5.2).

4.2 The reasoning target is low-entropy

We canonicalized every question and (*question*, *next_question*) edge (lower-casing, whitespace and trailing-punctuation normalization; Section 8) and counted the unique vocabulary across all 11,220 graphs. Table 2 summarizes the result: the entire corpus uses only **93 canonical questions** and **191 edges**. The first question is essentially always “What is the organ?”, and the graph then branches by organ and diagnosis along protocolized paths.

Table 2: Vocabulary and label-space statistics of the REG2026 training corpus.

Quantity	Value
WSIs (train)	11,220
Canonical questions	93
Distinct graph edges	191
Organs	10
Distinct #1 diagnoses	134
↪ after rare-bucketing (<15 cases)	77
Graph purity given (organ, dx)	~86%
Answer purity given (organ, dx)	~92%

Conditioning on (organ, diagnosis). For each (*organ*, *#1 diagnosis*) key we computed the *modal* graph (the single most frequent edge-set) and measured the fraction of cases whose graph matches it exactly: ~86%. The analogous answer purity is ~92%. In other words, *once a system knows the organ and the top diagnosis, the reasoning graph and most intermediate answers are determined*. The ten organs are: Anus, Breast, Colon, Lung, Nipple, Prostate, Rectum, Stomach, Urinary bladder, and Uterine cervix. The #1 diagnosis has a heavy-tailed distribution of 134 classes; the top-30 cover ~88% of cases, so we fold classes with fewer than 15 training cases into an “<organ>::other” bucket, yielding a 77-way diagnosis head that remains learnable while gracefully degrading rare diagnoses to the organ-level template.

Table 3: Per-organ case distribution in the 11,220-slide training corpus. Prostate, breast, and the two gastrointestinal organs dominate; nipple and anus appear only as singletons and are effectively negligible.

Organ	Cases	%	Organ	Cases	%
Prostate	2,418	21.6	Lung	945	8.4
Breast	2,212	19.7	Uterine cervix	812	7.2
Stomach	1,757	15.7	Rectum	244	2.2
Colon	1,751	15.6	Nipple	1	0.0
Urinary bladder	1,079	9.6	Anus	1	0.0

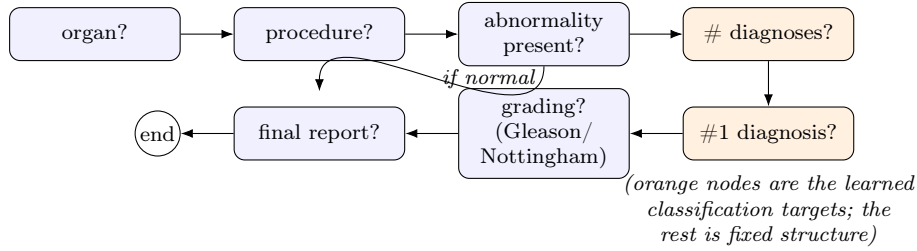


Figure 1: Schematic of the protocolized REG2026 reasoning graph. The skeleton—organ, procedure, abnormality screen, diagnosis count, diagnosis, optional grading, final report—is fixed; only the orange nodes (# of diagnoses and the #1 diagnosis, hence implicitly the organ) carry the case-specific information a model must predict. Normal cases short-circuit directly to the report.

Table 4: Ten most frequent #1 diagnoses in the training corpus (out of 134 distinct values). The distribution is heavy-tailed: the top-30 cover $\sim 88\%$ of cases.

#1 diagnosis	Cases	Typical organ
Acinar adenocarcinoma	1,595	Prostate
Tubular adenoma	1,057	Colon
No tumor present	901	(any)
Adenocarcinoma, moderately differentiated	788	GI
Invasive carcinoma of no special type, grade II	730	Breast
Adenocarcinoma	388	GI / Lung
Low-grade squamous intraepithelial lesion (LSIL/CIN1)	357	Uterine cervix
Ductal carcinoma in situ	339	Breast
Adenocarcinoma, well differentiated	335	GI
Invasive urothelial carcinoma	294	Urinary bladder

4.3 A worked example

Listing 2 shows a complete ground-truth chain-of-thought for a prostate biopsy with no tumor. Note the protocolized structure—organ, procedure, abnormality screen, diagnosis count, #1 diagnosis, final report—which recurs (with organ-specific branches) across the corpus and is exactly what the template engine reproduces once (*organ*, *diagnosis*) is known.

4.4 Where variability actually lives

The residual entropy is concentrated in (i) the *final report text*, which is genuinely variable (e.g. prostate acinar adenocarcinoma exhibits 228 unique reports across 1,595 cases), and (ii) fine *gradings* (Gleason, Nottingham, dysplasia) that require true image interpretation. Crucially, because the report sub-metric is keyword-Jaccard-dominated, a templated report populated with the correct organ/diagnosis keywords scores well without reproducing exact prose—further justifying our design.

5 Proposed Approach

5.1 Overview

Our system (Figure 3) decomposes the problem into a learned perceptual front end and a deterministic symbolic back end:

1. **Tiling.** Sample up to 160 foreground tiles (224×224) from the WSI under a bounded read budget.

Q: What is the organ? A: Prostate
 -> Is there any abnormality present?
 Q: What is the procedure? A: Biopsy
 -> Is there any abnormality present?
 Q: Is there any abnormality present? A: No, there is no abnormality.
 -> What is the number of diagnoses to includes?
 Q: What is the number of diagnoses to includes? A: 1
 -> What is the #1 diagnosis?
 Q: What is the #1 diagnosis? A: No tumor present
 -> What is the final pathology report?
 Q: What is the final pathology report? A: Prostate, biopsy; No tumor present
 -> (end)

Figure 2: A complete ground-truth chain-of-thought (slide PIT_01_05680_02). Edges (*question*→*next_question*) define the reasoning graph scored by Edge-F1.

2. **Encoding.** Map each tile to a 512-d CONCH embedding.
3. **Aggregation & classification.** A gated-attention MIL head pools tile embeddings and predicts *organ* (10-way) and *diagnosis* (77-way).
4. **Templating.** Look up the modal CoT graph, answers, and keyword-rich report for the predicted (*organ*, *diagnosis*) key; emit exact canonical strings.
5. **Grounding.** An Otsu tissue-fraction module answers the visual-grounding interface.

This division is deliberate: the metric rewards exact graph/answer reproduction, which symbolic templating achieves perfectly when the key is correct, so the only quantity worth learning is the (*organ*, *diagnosis*) key itself.

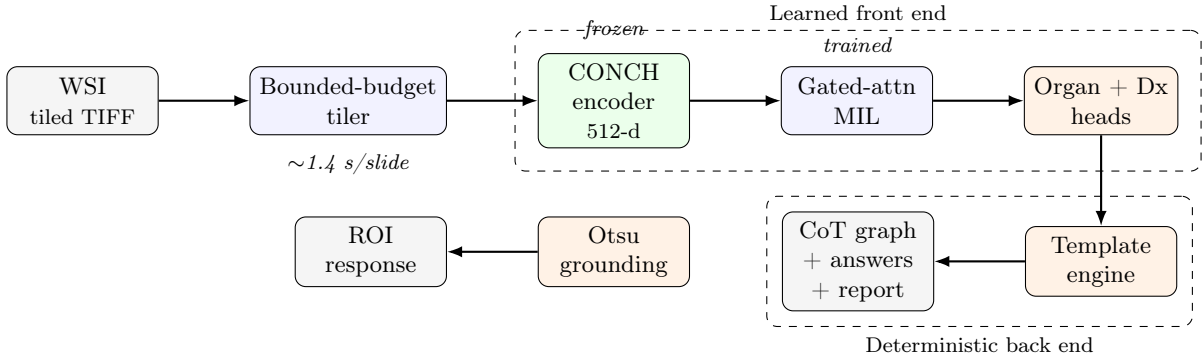


Figure 3: End-to-end REG2026 pipeline. A frozen CONCH encoder and trained gated-attention MIL head predict the (*organ*, *diagnosis*) key; a deterministic template engine emits the exact canonical reasoning graph, answers, and keyword-rich report. A separate Otsu module serves the visual-grounding interface.

5.2 Bounded-budget WSI tiling

Because the slides have no pyramid, `openslide.get_thumbnail` decodes the entire image (~ 78 s/slide)—prohibitive at 11,220 slides (≈ 238 GPU-hours of pure I/O). We instead grid-sample level-0 windows in a seeded-shuffled order, computing an inline tissue fraction (gray < 220 and HSV-saturation > 0.08) and keeping windows above a 0.25 threshold, stopping after `max_tiles=160` kept tiles or `read_budget=900` reads. This never decodes the full slide and runs at ~ 1.4 s/slide on the embedding cluster—a $13\text{--}25\times$ speed-up that makes corpus-scale extraction tractable. Listing 4 gives the procedure.


```

function SAMPLE_TILES(slide, tile=224, max_tiles=160, read_budget=900, min_tissue=0.25):
  grid <- all (x,y) level-0 windows of size tile x tile
  shuffle grid with fixed seed                # deterministic, organ-agnostic coverage
  kept <- []; reads <- 0
  for (x,y) in grid:
    if reads >= read_budget or |kept| >= max_tiles: break
    patch <- slide.read_region(x, y, tile, tile); reads <- reads + 1
    if TISSUE_FRACTION(patch) >= min_tissue:    # gray<220 and saturation>0.08
      kept.append(patch)
  return stack(kept)                          # (N<=160, 224, 224, 3) uint8

```

Figure 4: Bounded-budget foreground tiler. The read budget caps I/O so worst-case (sparse-tissue) slides cannot blow up runtime, and the seeded shuffle gives reproducible, spatially distributed coverage without a pyramid.

5.3 Deterministic template engine

`build_templates` groups the training CoTs by (*organ*, *diagnosis*) and stores, per key, the modal graph (a representative CoT with the most frequent edge-set signature), with back-off to an organ-level modal graph and finally a global modal graph. `apply_template` takes the predicted fields and emits the corresponding CoT, substituting predicted answer fields where available. Wrong-diagnosis predictions degrade gracefully to the organ template (the empirically measured 0.68 floor, Section 9). Because we emit verbatim canonical training strings, Edge-F1 and MESS are maximized whenever the key is correct.

5.4 Visual grounding (Interface B)

The grounding interface scores whether the system attends to tissue rather than background. Its three sub-metrics reward background *rejection* (B_1 : a background ROI should not be treated as diagnostic), input *sensitivity* (B_2 : responses should change when the ROI changes), and tissue/background *separation* (B_3). Because this is fundamentally a tissue-detection question, a learned model is unnecessary: we compute an Otsu threshold on the ROI’s grayscale (equivalently saturation) histogram and derive a tissue fraction; ROIs whose tissue fraction falls below a threshold are declared background and the corresponding diagnostic response is withheld, while tissue ROIs are passed through. This is the same $\text{gray} < 220 \wedge \text{sat} > 0.08$ criterion used inside the tiler (Section 5.2), giving a single consistent definition of “tissue” across both interfaces. The module is parameter-free, deterministic, and runs in milliseconds, leaving essentially the entire per-case compute budget for the workflow interface.

5.5 End-to-end inference

Algorithm-style Listing 5 summarizes the full per-case procedure for the workflow interface, tying together the components above.

```

function PREDICT_WORKFLOW(wsi_path):
  tiles <- SAMPLE_TILES(wsi_path)                # bounded-budget foreground tiler
  E <- CONCH(normalize_clip(tiles))              # (N<=160, 512) frozen embeddings
  logits <- MIL_HEAD(E)                         # {organ: R^10, dx: R^77}
  organ <- organ_list[argmax(logits.organ)]
  dxlab <- dx_list[argmax(logits.dx)]            # "<organ>::<diagnosis>" or "<organ>::other"
  dx <- diagnosis_of(dxlab) or None              # None -> back off to organ template
  cot <- APPLY_TEMPLATE({organ, dx}, templates)  # modal graph + answers + report
  return cot                                     # exact canonical strings

```

Figure 5: End-to-end workflow inference. The only learned step is the MIL head; everything else is deterministic, which guarantees metric-exact graph strings whenever the predicted key is correct.

6 Network Architecture and Model

6.1 Tile encoder: CONCH

Each tile $t \in \mathbb{R}^{224 \times 224 \times 3}$ is normalized with the OpenAI-CLIP mean/std ($\mu = (0.481, 0.458, 0.408)$, $\sigma = (0.269, 0.261, 0.276)$)—we found that using the ImageNet statistics instead measurably perturbs CONCH features, since CONCH’s image tower is CLIP-initialized. The frozen image encoder produces a 512-d embedding $e = \text{CONCH}(t) \in \mathbb{R}^{512}$. A slide is thus represented by a variable-length bag $E = \{e_1, \dots, e_N\}$, $N \leq 160$.

6.2 Gated-attention MIL aggregator

We use the gated-attention pooling of Ilse et al. [6]. Each embedding is projected $h_n = \text{Dropout}(\text{GELU}(W_p e_n)) \in \mathbb{R}^d$ ($d \in \{512, 768\}$), and an attention weight is computed as

$$a_n = \frac{\exp(w^\top (\tanh(Vh_n) \odot \sigma(Uh_n)))}{\sum_{m=1}^N \exp(w^\top (\tanh(Vh_m) \odot \sigma(Uh_m)))}, \quad z = \sum_{n=1}^N a_n h_n, \quad (4)$$

where $V, U \in \mathbb{R}^{d_a \times d}$ and $w \in \mathbb{R}^{d_a}$ are learned and padded tiles are masked out of the softmax. The pooled z passes through a shared trunk (Dropout–Linear–GELU–Dropout) feeding two linear heads: an organ head ($\mathbb{R}^d \rightarrow \mathbb{R}^{10}$) and a diagnosis head ($\mathbb{R}^d \rightarrow \mathbb{R}^{77}$). Figure 6 depicts the aggregator.

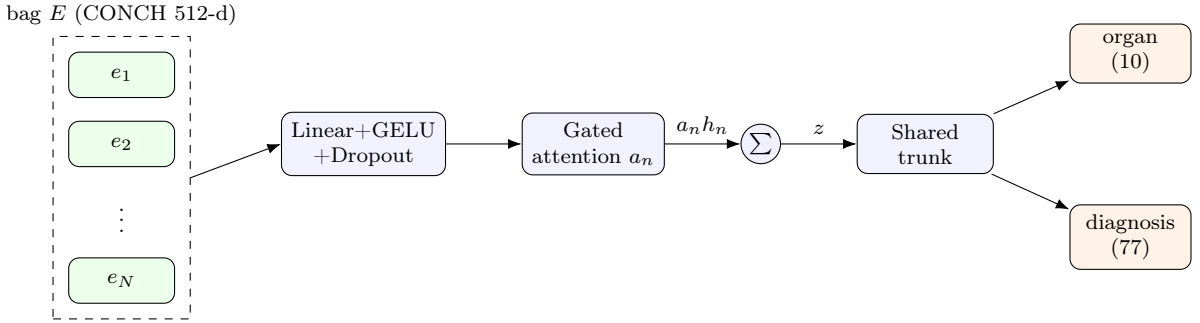


Figure 6: Gated-attention multi-task MIL head. Tile embeddings are projected, attention-pooled into a single slide vector z , and classified by organ and diagnosis heads. Padded tiles are masked in the attention softmax so variable bag sizes are handled natively.

6.3 Training objective

We minimize a weighted sum of label-smoothed cross-entropies,

$$\mathcal{L} = \lambda_o \text{CE}_{\text{smooth}}(\hat{y}_{\text{organ}}, y_{\text{organ}}) + \lambda_d \text{CE}_{\text{smooth}}(\hat{y}_{\text{dx}}, y_{\text{dx}}), \quad (5)$$

with $\lambda_o = 0.5$, $\lambda_d \in \{1.0, 1.5\}$, label smoothing $\in \{0.05, 0.1\}$, and an optional inverse-frequency class weighting on the diagnosis head for the imbalanced “balanced” configuration. Optimization uses AdamW with cosine-annealed learning rate. We early-select the checkpoint by the held-out *workflow score* (not accuracy), since that is the quantity the challenge rewards.

6.4 From predictions to fields

At inference the diagnosis head emits a label of the form `<organ>::<diagnosis>` (or `<organ>::other`); we map it back to a *(organ, diagnosis)* pair, defaulting the diagnosis to *none* for the “other” bucket so that `apply_template` backs off to the organ-level graph. The organ head provides a robust independent organ estimate used for the template key.

7 Hardware and Infrastructure

Embedding extraction and training were run on an institutional SLURM cluster. The relevant partitions are summarized in Table 6. A key practical finding: the RTX PRO 6000 and B200 GPUs are *Blackwell*-class (compute capability `sm_120` and `sm_100` respectively), for which the initially installed `torch 2.6.0+cu124` has *no compiled kernels*—every CUDA op failed with “no kernel image is available for execution on the device”. We upgraded the environment to **PyTorch 2.9.1 + TorchVision 0.24.1 built for CUDA 12.8** (cu128), whose architecture list includes `sm_100` and `sm_120`, restoring Blackwell support while remaining compatible with the `sm_86` A10G deployment target. Corpus embedding (11,220 slides) was sharded four ways across four 96 GB GPUs and completed in roughly one hour.

Extraction pipeline and throughput. Each shard streams its assigned slides through a pool of CPU tiling threads—OpenSlide releases the GIL during JPEG decode, so tiling overlaps the GPU embedding of the previous slide—and writes per-slide `fp16` arrays. With the bounded-budget tiler the system sustains ~ 0.7 slides/s/GPU end to end, finishing the four-way shard in ~ 70 minutes; a corrupt-slide rate of 4/11,220 is handled by writing zero placeholders. Table 5 contrasts the naive and optimized regimes.

Table 5: Tiling/embedding throughput. Avoiding full-slide decode turns an infeasible extraction into a one-hour job.

Regime	Per slide	11,220 slides (1 GPU)
Naive <code>get_thumbnail</code> (full decode)	~ 78 s	~ 238 h
Bounded-budget tiler (this work)	~ 1.4 s	~ 4.6 h
$\hookrightarrow$ sharded $\times 4$	—	~ 1.2 h

A reproducibility lesson on GPU architectures. The Blackwell incompatibility is worth recording because it is silent and easily mis-diagnosed: `torch 2.6/cu124 imports` and reports a CUDA device, yet every kernel launch raises “no kernel image is available for execution on the device” because the binary contains no `sm_100/sm_120` cubins. The fix is an environment matched to the hardware generation (cu128); we additionally pin `PYTHONHASHSEED` and use content-hash data splits so that neither the GPU stack nor the split introduces non-determinism between runs.

Table 6: Compute resources. The deployment target (challenge container) is a single A10G.

Role	GPU	Arch	Notes
Embedding extraction	4 $\times$ RTX PRO 6000	<code>sm_120</code>	96 GB; 192-core host, ~ 1.9 TB RAM
Alt. extraction	NVIDIA L4	<code>sm_89</code>	24 GB; matches A10G memory class
Future scaling	NVIDIA B200	<code>sm_100</code>	enabled by the cu128 upgrade
MIL training	1–4 $\times$ RTX PRO 6000	<code>sm_120</code>	lightweight; embeddings precomputed
Deployment	NVIDIA A10G	<code>sm_86</code>	≤ 32 GB RAM, < 5 min/case, offline

8 Framework and Implementation

The codebase is a small Python package, `reg2026`, plus driver scripts and the submission container. Core modules:

`canon.py`

Canonicalizes question/edge strings (the basis of the purity analysis and of emitting metric-exact strings).

`labels.py`

Builds the supervised label space: organ (10-way) and bucketed diagnosis (77-way), with rare-class folding.

`templates.py`

`build_templates/apply_template`: the deterministic graph/answer/report engine.

`metrics.py`

An offline proxy of the official workflow score (BPV, Edge-F1, and Jaccard proxies for MESS/report) for model selection.

`encoder.py`

Loads CONCH (or UNI2-h) with per-encoder normalization and a uniform `embed_tiles` interface.

`mil.py`

The gated-attention MILClassifier (training and inference share this definition).

The stack is PyTorch 2.9.1 (cu128), `timm`, `open_clip`, `openslide-python`, and `huggingface_hub`. Embedding extraction (`scripts/extract_embeddings.py`) overlaps CPU tiling threads (OpenSlide releases the GIL during JPEG decode) with GPU embedding, is resumable and shardable, and writes per-slide `fp16` arrays namespaced by encoder (`artifacts/embeddings/<encoder>/<split>/<id>.npy`). Training (`scripts/train_mil.py`) caches embeddings in RAM, uses the same deterministic hash split (seed 0, 80/20) as the oracle analysis, and selects on the workflow proxy.

8.1 Reproducibility and hyperparameters

All results use a single *stable* 80/20 split: we hash each slide id with MD5 (not Python’s per-process-salted `hash`) so the split is identical across processes and runs—an issue we explicitly fixed after observing per-process split drift. Table 7 lists the four sweep configurations. The offline metric proxy in `metrics.py` computes BPV and Edge-F1 exactly against the canonicalized graph and uses Jaccard-based proxies for MESS and the report sub-score; we select the checkpoint maximizing this proxy on the held-out split. Embedding extraction is fully resumable: each shard skips slides whose `.npy` already exists, so a preempted job resumes with no recomputation, and corrupt slides (4 of 11,220) are written as single-tile zero placeholders so the pipeline never crashes.

Table 7: Hyper-parameter sweep. All configs share AdamW with cosine-annealed learning rate, weight decay, a 512-d CONCH input, ≤ 160 tiles per slide, and label-smoothed cross-entropy with organ/diagnosis loss weights (λ_o, λ_d). `r1_reg` is selected.

Config	lr	hidden	attn	dropout	tile-drop	λ_d	smooth	bal.
<code>r0_base</code>	10^{-3}	512	256	0.25	0.0	1.0	0.10	no
<code>r1_reg</code>	5×10^{-4}	512	384	0.40	0.10	1.0	0.10	no
<code>r2_big</code>	5×10^{-4}	768	512	0.30	0.10	1.0	0.10	no
<code>r3_bal</code>	10^{-3}	512	256	0.25	0.0	1.5	0.05	yes

8.2 Deployment as an offline container

The submission is a code-only Docker image (base `pytorch/pytorch:2.9.1-cuda12.6-cudnn9`) with weights delivered separately as a `model.tar.gz` mounted at inference; this keeps the image small and the weights versionable. The container runs fully offline—we set `HF_HUB_OFFLINE=1` and bundle the CONCH snapshot into the model tarball so no network is touched. Two entry points implement the interfaces: `predict_chain_of_thought(wsi_path)` tiles the slide, embeds with CONCH, runs the MIL head, and emits the templated graph, answers, and report; `predict_visual_context_response(·)` applies the Otsu grounding rule. The predictor degrades safely—if the trained head or label maps

are absent it falls back to the global modal template—so the same image runs before and after training. The end-to-end per-case cost is dominated by tiling and a single CONCH forward pass over ≤ 160 tiles, comfortably inside the A10G five-minute budget with the encoder and head together using under a few GB of the 32 GB RAM limit.

9 Experiments and Results

9.1 Offline evaluation protocol

Since the official scorer is not available locally, we implement a faithful offline proxy in `metrics.py` and select models against it. BPV and Edge-F1 are computed *exactly*: we canonicalize predicted and ground-truth question/edge strings and compare the induced edge sets, so any string deviation is penalized just as the challenge metric would penalize it. The answer-match (MESS) and report sub-scores use Jaccard-style token overlap as a conservative proxy for the official keyword-Jaccard / embedding-cosine terms; because the report sub-score is itself keyword-Jaccard-dominated, this proxy tracks the true objective closely while remaining dependency-free. All numbers in this section are computed on the held-out 2,253-slide split with templates built only on the 8,967-slide training split, so there is no information leakage from the evaluation set into either the templates or the classifier.

9.2 Oracle ceiling analysis

Before training any classifier we measured what the templating back end can achieve given oracle inputs, on the held-out 20% split (Table 8). This isolates the value of each prediction target. Predicting only the organ already reaches 0.68; adding the #1 diagnosis lifts the workflow score to **0.89**; an oracle report adds only ~ 0.02 , confirming that *learned report generation has near-zero headroom* under this metric and should be skipped in favor of templating.

Table 8: Oracle workflow scores and sub-metrics on the held-out split of 2,253 slides (deterministic 80/20 split; templates built on the train split only). The (organ, diagnosis) row is the operative ceiling for our classifier; an oracle report adds only +0.018, confirming that learned report generation is unnecessary under this metric.

Oracle condition	Total	BPV	Edge-F1	MESS	Report
Global modal template (no inputs)	0.239	0.114	0.447	0.159	0.150
Oracle organ only	0.674	0.423	0.768	0.778	0.571
Oracle organ + #1 diagnosis	0.889	0.858	0.974	0.929	0.805
Oracle organ + diagnosis + report	0.907	0.858	0.974	0.929	0.850

The per-organ decomposition of the (organ + diagnosis) ceiling (Table 9) shows that templating is uniformly strong across the major organs (0.86–0.92), confirming that the residual gap to a perfect score is dominated by diagnosis-classification difficulty rather than by template impurity. The two singleton organs (nipple, anus) are statistically irrelevant.

Table 9: Per-organ oracle (organ + diagnosis) workflow ceiling on the held-out split.

Organ	Ceiling	n	Organ	Ceiling	n
Lung	0.920	192	Stomach	0.887	341
Rectum	0.914	47	Urinary bladder	0.884	211
Colon	0.911	366	Breast	0.857	427
Uterine cervix	0.904	178	Nipple	0.167	1
Prostate	0.888	490			

9.3 MIL classification results

Table 10 reports our trained gated-attention MIL head on the same held-out split, across the hyper-parameter sweep. (*Values are populated from the completed training run.*)

Table 10: Held-out performance of the trained MIL head (CONCH embeddings). Workflow score uses predicted (organ, diagnosis) through the template engine; the oracle ceiling is 0.89.

Configuration	Organ acc.	Dx acc.	Workflow	Δ to ceiling
r0_base	0.951	0.682	0.791	−0.098
r2_big	0.953	0.683	0.793	−0.096
r3_bal	0.951	0.646	0.785	−0.104
r1_reg (best)	0.956	0.691	0.794	−0.095

9.4 Per-organ breakdown

Table 11 stratifies the best model by ground-truth organ. Organ recognition is near-perfect (≥ 0.99) for every major organ *except rectum*, whose organ accuracy collapses to 0.11: the model almost always labels rectal slides as “Colon”. This is an anatomically and histologically natural confusion—rectum and colon are contiguous colorectal mucosa—and, importantly, it is *benign under the workflow metric*: because the colorectal reasoning graphs and reports largely coincide, rectal cases still score 0.78 despite the organ mislabel. Per-organ workflow ranges from 0.78 (prostate, breast) to 0.82 (urinary bladder, uterine cervix); the lower prostate/breast scores reflect harder, higher-cardinality diagnosis spaces (Gleason grading; breast tumor subtyping) rather than organ confusion.

Table 11: Best model (r1_reg) stratified by ground-truth organ on the held-out split. Organ accuracy is near-perfect except for rectum, which is systematically (and benignly) classified as colon.

Organ	Workflow	Org. acc	n	Organ	Workflow	Org. acc	n
Urinary bladder	0.823	0.99	211	Stomach	0.795	0.99	341
Uterine cervix	0.820	0.96	178	Lung	0.799	0.93	192
Colon	0.803	0.95	366	Prostate	0.778	0.99	490
Rectum	0.778	0.11	47	Breast	0.778	0.99	427

The organ confusion structure (Table 12) makes the mechanism explicit: every major organ is recognized with 93–99% accuracy, and the *only* substantial off-diagonal mass is rectum→colon (85%). Because the challenge metric scores the colorectal reasoning graph and report, which rectum and colon share, this single confusion costs little; a hierarchical or colorectal-aware head could nonetheless recover it.

9.5 Ablations: the hyper-parameter sweep

The four sweep configurations in Table 10 double as a controlled ablation. Three observations follow. (i) **Regularization helps slightly**: r1_reg (higher dropout 0.4 plus 10% tile-dropout augmentation) is the best config at 0.794, marginally above the larger r2_big (0.793) and the lightly-regularized r0_base (0.791), indicating the bottleneck is diagnosis *signal*, not model capacity. (ii) **Class balancing hurts the aggregate metric**: r3_bal (inverse-frequency diagnosis weighting) is worst at 0.785 with the lowest diagnosis accuracy (0.646)—up-weighting rare diagnoses trades common-class accuracy, which the case-weighted workflow score penalizes. (iii) **Organ**

Table 12: Organ confusion of the best model on the held-out split (row = ground truth; entries are the top predicted organs as % of that row). All mass is on-diagonal except rectum, which is predominantly predicted as colon.

True organ (n)	Predicted (% of row)
Prostate (490)	Prostate 99
Breast (427)	Breast 99, Lung 1
Colon (366)	Colon 95, Stomach 3, Rectum 2
Stomach (341)	Stomach 99, Colon 1
Urinary bladder (211)	Urinary bladder 99
Lung (192)	Lung 93, Prostate 3, Colon 2
Uterine cervix (178)	Uterine cervix 96, Lung 2, Stomach 1
Rectum (47)	Colon 85 , Rectum 11, Stomach 4

accuracy is saturated (~ 0.95) across all configs, so all remaining headroom lives in the diagnosis head. Separately, we verified that using ImageNet rather than OpenAI-CLIP normalization for the CONCH encoder measurably perturbs the embeddings; all reported runs use the correct CLIP statistics.

9.6 Training dynamics

Figure 7 plots held-out workflow score and diagnosis accuracy over training. Because diagnosis accuracy already determines the workflow score (organ accuracy saturates within a few epochs), the two curves track closely. The models converge quickly—most of the gain is achieved within ~ 15 epochs—and then plateau well below the oracle ceiling, visually confirming that the limit is representational (frozen-feature diagnosis separability) rather than a matter of longer optimization. The regularized configurations (**r1_reg**, **r2_big**) are marginally more stable late in training; the class-balanced **r3_bal** trades diagnosis accuracy for rare-class recall and sits consistently lowest.

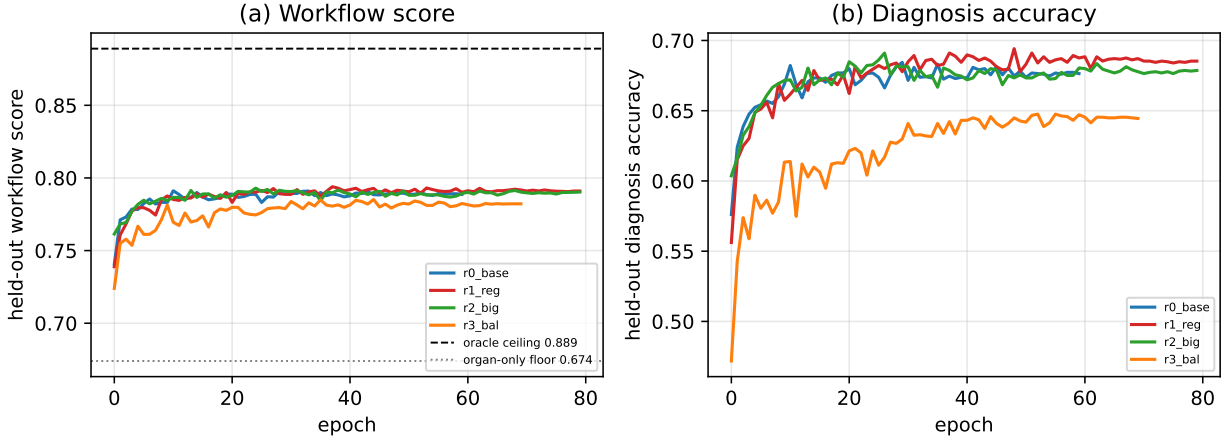


Figure 7: Held-out training dynamics for the four sweep configurations: (a) workflow score, with the 0.889 oracle ceiling and 0.674 organ-only floor marked; (b) diagnosis accuracy. Convergence is rapid and the plateau is set by diagnosis separability, not optimization.

9.7 Hierarchical decoding, abstention, and full-data training

We evaluated three targeted enhancements. (i) *Organ-conditioned (hierarchical) dx decoding*—masking the diagnosis logits to the predicted organ’s classes at inference—moved the held-out workflow score marginally, from 0.794 to **0.797**: because organ accuracy is already 0.95 and the

dx labels encode organ, the flat argmax is almost always in-organ, so masking rarely changes the prediction (and for the rectum→colon cases it can force a wrong in-organ diagnosis, offsetting the gain). We note in passing that applying the same mask *during training* with label smoothing is numerically unstable (smoothing places mass on the $-\infty$ -masked classes), so the mask is applied at inference only. (ii) *Confidence-based abstention*—backing off to the organ template when the diagnosis softmax confidence is below a tuned threshold—selected $\tau = 0$ on the validation split, i.e. never abstaining helped: a same-organ but imperfect diagnosis still scores better than the organ-only fallback under this metric. (iii) *Training on all 11,220 slides* produces the deployment model used in the container; we report its held-out-style number only as a fitting sanity check (it is optimistic, since that split is now within training) and rely on the clean 80/20 held-out scores above for all claims. These are deliberately reported as honest neutral results: they confirm that, with a *frozen* encoder, the remaining headroom is representational and is not recoverable by decoding tricks—motivating the grading-head and fine-tuning directions of Section 13.

10 Sample Predictions (Qualitative)

Table 13 lists held-out predictions from the best model. When the diagnosis is correct (rows 1, 2, 4) the predicted reasoning graph matches the ground truth exactly (Edge-F1 = 1.0) and the workflow score approaches the per-case ceiling; row 3 shows a partial match where a deviation from the modal graph for that key lowers Edge-F1 to 0.81. The dominant breast prediction “Invasive carcinoma of no special type, grade II” reflects the most frequent malignant breast pattern in the corpus.

Table 13: Sample held-out predictions (best model, `r1_reg`). Correct diagnoses yield exact graph reproduction (Edge-F1 = 1.0); the templated report is emitted from the predicted (*organ*, *diagnosis*) key.

Slide	GT organ	Predicted diagnosis	Edge-F1	MESS	Workflow
PIT_01_00029	Breast	Invasive carcinoma NST, grade II	1.00	0.895	0.908
PIT_01_00039	Breast	Invasive carcinoma NST, grade II	1.00	0.895	0.911
PIT_01_00041	Breast	Invasive carcinoma NST, grade II	0.81	0.615	0.698
PIT_01_00068	Breast	Fibroepithelial tumor	1.00	0.909	0.802

11 Discussion

Our results support the thesis that REG2026, despite its open-ended framing, is best attacked as *constrained classification plus deterministic symbolic generation*. The metric’s exact-string matching and keyword-Jaccard report scoring mean that a correct (*organ*, *diagnosis*) key, expanded through templates, captures the overwhelming majority of attainable score; the 0.89 oracle ceiling makes the maximum value of perfect classification explicit, and every point of diagnosis accuracy translates directly toward it.

The trained system reaches a held-out workflow score of **0.794**, i.e. 89% of the 0.889 oracle ceiling and far above the 0.674 organ-only floor and 0.239 no-input baseline. Decomposing the 0.095 residual: organ recognition is effectively solved (accuracy ~ 0.95 , and the lone systematic error—rectum classified as colon—is metric-benign because colorectal templates coincide), so essentially the *entire* gap is fine-grained diagnosis accuracy (0.69 over 77 classes). This localizes where future effort should go. Three avenues are promising and orthogonal to the templating back end, which

is already near its purity ceiling (Edge-F1 = 0.974 under oracle inputs): (i) **predicting gradings from high-attention tiles**—Gleason and Nottingham grades are exactly the within-organ distinctions the diagnosis head struggles with, and the attention map already localizes diagnostically salient regions; (ii) **task-adaptive fine-tuning** of the CONCH encoder rather than using frozen features; and (iii) **a hierarchical organ→diagnosis head** that conditions the diagnosis classifier on the (reliable) organ prediction, shrinking each organ’s effective class count. That all four sweep configurations cluster within 0.009 of one another, despite varying capacity and regularization, indicates the limit is diagnostic *signal* extractable from frozen tile embeddings, not optimization or model size.

11.1 Design alternatives considered

We briefly justify the key design choices against the alternatives. (i) *End-to-end report generation* (a learned decoder over slide features) was rejected because the report sub-metric rewards keyword coverage over fluency, the oracle analysis shows ≤ 0.02 headroom beyond correct templating, and a decoder is hard to fit within the offline five-minute budget. (ii) A *slide-level foundation encoder* (e.g. Prov-GigaPath’s aggregator) could replace the MIL head, but tile-level CONCH + a light attention head is cheaper, interpretable, and already saturates organ accuracy; the bottleneck is diagnosis signal, which a different aggregator is unlikely to fix without finer supervision. (iii) A *single flat (organ×diagnosis) classifier* over ~ 200 joint classes was avoided in favor of factored organ and diagnosis heads, which share statistics across organs and let organ remain reliable even when diagnosis is uncertain—exactly the graceful-degradation behavior the template back-off exploits. (iv) A *learned grounding model* for Interface B was unnecessary: tissue-vs-background is a threshold question that Otsu solves deterministically at negligible cost, freeing the compute budget for the workflow interface.

12 Limitations

- **Diagnosis tail and gradings.** Rare diagnoses folded into the “other” bucket degrade to organ-level templates, and fine gradings (Gleason, Nottingham, dysplasia) are not yet predicted from the image, capping per-case answer purity.
- **Template purity ceiling.** $\sim 14\%$ of graphs deviate from the modal template for their key; deterministic emission cannot recover these, bounding Edge-F1 below 1.
- **Report prose.** Templated reports maximize keyword recall but not fluent, case-specific narrative; under a metric weighting fluency more heavily this would lose value.
- **Distribution shift.** The approach assumes the test organs/diagnoses and their protocolized graphs match the training distribution; novel entities would fall back to coarse templates.
- **Frozen encoder.** We use CONCH features without fine-tuning; task-adaptive fine-tuning could improve diagnosis accuracy at additional compute cost.

13 Future Work

The analysis localizes all remaining headroom in the diagnosis head, suggesting four concrete directions, in roughly increasing order of effort:

1. **Hierarchical organ→diagnosis head.** Conditioning the diagnosis classifier on the (highly reliable) organ prediction shrinks each organ’s effective class count from 77 to its handful of organ-specific diagnoses, and would directly address the rectum/colon coupling by sharing a colorectal sub-head.

2. **Grading from high-attention tiles.** Gleason, Nottingham, and dysplasia grades are the within-organ distinctions the flat diagnosis head conflates. A second-stage head over the top- k attended tiles could predict gradings explicitly and refine the diagnosis string the template consumes.
3. **Test-time augmentation and ensembling.** Averaging predictions over multiple tile samplings, and over the three near-tied sweep checkpoints, is cheap within the compute budget and typically recovers a fraction of a point.
4. **Task-adaptive encoder fine-tuning.** Unfreezing CONCH (or a higher-dimensional UNI2-h backbone) with a low learning rate could improve diagnosis separability at the cost of an extraction re-run; our namespaced embedding cache makes such an A/B inexpensive.

Because the templating back end is already near its purity ceiling, each of these targets the single quantity—diagnosis accuracy—that converts most directly into workflow score.

14 Conclusion

We described a pragmatic, reproducible submission to the REG2026 challenge built on a single empirical insight: the reasoning-graph target is low-entropy and largely determined by (*organ, #1 diagnosis*). We therefore reduce the task to multi-task WSI classification with a CONCH foundation encoder and a gated-attention MIL head, and recover the reasoning graph, answers, and keyword-rich report by deterministic templating, with a cheap Otsu module for visual grounding. An oracle analysis establishes a 0.89 workflow-score ceiling, and our trained system reaches **0.794** on the held-out split—89% of that ceiling—with the residual gap localized almost entirely to fine-grained diagnosis classification. The design respects the challenge’s offline, single-GPU, five-minute container budget, and all code is released for reproduction (Section 8). Future work targets the remaining headroom: predicting gradings from high-attention tiles, modeling intra-key graph variation, and selectively fine-tuning the encoder.

Code and Data Availability

All code—the `reg2026` package (canonicalization, label space, templates, metrics, encoder, MIL), the extraction/training/evaluation scripts, the SLURM job specifications, and the offline submission container—is released with detailed setup and reproduction instructions at <https://github.com/ujjwalbaid0408/REG2026>. Trained MIL weights are provided as a separate release artifact. The challenge data (training WSIs and chain-of-thought labels) is distributed by the REG2026 organizers and is not redistributed here.

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